
MARIA AYDIN

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Master's student in Computer Science specializing in *Computer Vision and Machine Intelligence (VMI)* at Université Paris Cité, seeking a **4- to 6-month internship starting in February 2027**. Passionate about Computer Vision and Artificial Intelligence, I am seeking an internship in computer vision and deep learning to apply my skills to real-world engineering problems in a professional setting

FORMATION

UNIVERSITÉ PARIS CITÉ, PARIS, FRANCE — MASTER OF SCIENCE (M.SC.) IN COMPUTER SCIENCE — VISION & MACHINE INTELLIGENCE (VMI) - 2025-PRESENT

Core Modules : **Computer Vision & AI** (Image Analysis, Pattern Recognition, Neural Networks for Computer Vision, Machine Learning), **Processing & Data** (Signal & Image Processing, Big Data & Data Science), **Foundations & Algorithms** (Advanced Programming (Python, C++), Optimization & Algorithms, Computational Geometry, Probability & Statistics), **Cross-Disciplinary:** (Unix/Linux System Administration, Web Programming, Distributed Systems).

UNIVERSITÉ PARIS CITÉ, PARIS, FRANCE — BACHELOR OF SCIENCE (B.SC.) IN COMPUTER SCIENCE AND APPLICATIONS— 2022-2025

ACADEMIC & TECHNICAL PROJECTS

2025-2026: Weakly Supervised Learning for Histological Image Analysis and Detection

- Designed a weakly supervised pipeline to classify Whole Slide Images (WSI) into healthy vs. tumorous tissues without patch-level annotations.
- Encoded patches using DINOv3, generated 1D random walks (with backtracking) for DistilBERT sequence modeling, and created interactive attention maps.
- Skills & Stack:** Multimodal Models (Vision + NLP), Python, PyTorch, DINOv3, DistilBERT, Fine-Tuning, Hyperparameter Optimization, Interactive Visualization.

2025-2026: Information Retrieval Engine - Stack Exchange Document Ranking - M1 VMI Kaggle Competition

- Engineered a hybrid search engine combining BM25+ (lexical) and Sentence Transformers (semantic) via Reciprocal Rank Fusion (RRF) over 216k+ documents.
- Developed a query classifier (TF-IDF + SVM) with query expansion and conducted benchmark evaluations (Precision, Recall, MRR).
- Skills & Stack:** Sentence Transformers, FAISS, scikit-learn, Python.

2025-2026: Scalable and Robust Music Identification with Audio Embeddings, Vector Search and Fingerprinting

- Architected a real-time, scalable audio identification system operating on short snippets and mic inputs.
- Built a two-stage retrieval pipeline: k-NN vector search on embeddings (CLAP, MERT) followed by Shazam-style fingerprint reranking.
- Enhanced noise robustness using Room Impulse Response (RIR) acoustic data augmentation.
- Skills & Stack:** ChromaDB, FAISS, Audio Signal Processing (MFCC), CLAP.

2025-2026: Euro Coin Detection, Identification & Counting

- Developed an automated pipeline to detect, classify, and count Euro coins under occlusion and non-uniform lighting.
- Fused Hough Circle Transform, Watershed segmentation, and HSV color space classification.
- Evaluated performance quantitatively across 106 test images using regression metrics (MAE, RMSE, MAPE) and optimized Hough parameters via grid search.
- Skills & Stack:** Python, Classical Computer Vision (Hough, Watershed, CLAHE, Morphology), Color Analysis (HSV), OpenCV, NumPy, Pandas.

2025-2026: Shape Classification via Spectral & Geometric Descriptors

- Conducted a comparative study of 5 shape descriptors (E34, GFD, SA, F0, F2) on a dataset of 99 binary shapes across 9 classes facing occlusions, distortions, and intra-class variance.
- Implemented 3 machine learning strategies (k-NN, k-Means, Ensemble Majority Voting) evaluated via confusion matrices and PR/AUC curves.
- Skills & Stack:** Supervised & Unsupervised Machine Learning, Shape Descriptors, Model Evaluation, Python (NumPy, scikit-learn).

TECHNICAL & PERSONAL SKILLS

- Programming Languages:** Python (Advanced), C++, C, Java
- AI & Deep Learning:** PyTorch, TensorFlow, Hugging Face (Transformers), scikit-learn, Multimodal Models (DINOv3, CLAP, Vision-Language)
- Computer Vision & Signal Processing:** OpenCV, Image Segmentation, Feature Extraction, Classification, Audio Signal Processing (MFCC, Spectral Fingerprinting)
- Data, Databases & Versioning:** SQL, MySQL, Vector Search (FAISS, ChromaDB), SQLite, Parquet, Lexical & Semantic Models, Git, SVN
- Languages:** French (Native), English (Fluent)